

Occurrence of estrogen hormones in biosolids, animal manure and mushroom compost.

The presence of natural estrogen hormones as trace concentrations in the environment has been reported by many researchers and is of growing concern due to its possible adverse effects on the ecosystem. In this study, municipal biosolids, poultry manure (PM) and cow manure (CM), and spent mushroom compost (SMC) were analyzed for the presence of seven estrogen hormones. 17α -estradiol, 17β -estradiol, 17α -dihydroequilin, and estrone were detected in the sampled biosolids and manures at concentrations ranging from 6 to 462 ng/g of dry solids. 17α -estradiol, 17β -estradiol, and estrone were also detected in SMC at concentrations ranging from 4 to 28 ng/g of dry solids. Desorption experiments were simulated in the laboratory using deionized water (milli-Q), and the aqueous phase was examined for the presence of estrogen hormones to determine their desorption potential. Very low desorption of 0.4% and 0.2% estrogen hormones was observed from municipal biosolids and SMC, respectively. An estimate of total estrogen contribution from different solid waste sources is reported. Animal manures (PM and CM) contribute to a significant load of estrogen hormones in the natural environment.